# Why recognition passed while the ORB gate failed

Interpretation of the paused OLED component-test results
26 October 2026

**Main finding:** The OLED was recognized through ORB + RANSAC. The displayed 2.96% is a reference-feature match yield, not a 2.96% probability that the object is an OLED.

## 1. What the recorded frame contains

| Measurement | Recorded value | Meaning |
|---|---|---|
| Reference keypoints | 2,400 | Features available in the selected prepared reference. |
| Live keypoints | 962 | Unique camera-frame features reported by telemetry. |
| Good / attempted matches | 71 / 2,400 | Usable correspondences versus attempted descriptor queries. |
| Geometric inliers | 45 | Matches agreeing with the estimated geometric mapping. |
| Verification source | ORB + RANSAC | Photo-feature recognition followed by geometric verification. |
| Pose / orientation | Correct / Forward | The measured pose passed the implemented target-orientation check. |

## 2. How the recognition decision was reached

Camera image > preprocessing and ORB extraction > descriptor matching > distinctive correspondence filtering > RANSAC geometric consensus > boundary and applicable pose checks > verified target.

A good descriptor match must satisfy the shared rule **d1 < 0.75 x d2** and **d1 <= 64 Hamming distance**, with duplicate correspondences removed. RANSAC then checks whether the surviving points agree on one plausible transformation. An outline alone does not supply that evidence.

Scope: explanation of the supplied screenshot and the inspected current implementation. This is a read-only report; no application code, thresholds, references or saved progress were changed.

# Understanding the three percentages

## 3. Match yield: 2.96%

### 71 good matches / 2,400 attempted queries x 100 = 2.96%

The denominator counts attempted reference-descriptor queries. It does not count pixels, complete objects, recognition trials, or all possible descriptor-pair comparisons. Here it equals the reference keypoint count; that equality should not be assumed for every other matcher or ROI.

Many stored features may not produce a distinctive live correspondence because of viewpoint, scale, shadow, glare, blur, occlusion or repeated texture. The remaining 2,329 attempts did not survive the matching filters; that does not mean 2,329 confirmed observations of a wrong component.

## 4. Geometric consensus: approximately 63.4%

### 45 RANSAC inliers / 71 good matches x 100 = 63.38%

This asks how many surviving correspondences agree on the same geometry. It is separate from match yield. The production recognizer checks absolute match/inlier support, consensus, projected geometry and applicable pose requirements. It does not require 85% of all reference queries to match.

The default minimums in the inspected configuration are 12 good matches, 8 inliers and a 0.45 inlier fraction; reference-specific rules and additional checks can apply. Your screenshot exceeds those default count and consensus minimums. The screenshot does not expose every intermediate acceptance check.

## 5. Verifier confidence: 67.75%

This is a weighted heuristic score, not an independently calibrated probability of correct identification. The inspected standard candidate score combines inlier fraction (55%), a capped inlier-count score (25%) and descriptor-distance quality (20%). It is neither 71/2,400 nor 45/71.

## 6. Why PASS and FAIL can coexist

| Decision | Current interpretation |
|---|---|
| Recognition: Match (VERIFIED) | Enough spatially consistent evidence supported the selected OLED and its applicable validation checks. |
| Academic ORB gate: FAIL | The separate match-yield comparison is 2.96% against a fixed 85% threshold. That criterion failed. |

**This was not a shape-fallback rescue.** The displayed verification source is ORB + RANSAC. The recognition pipeline accepted its geometric evidence while the separate academic yield criterion failed.

# Interpreting the paused test

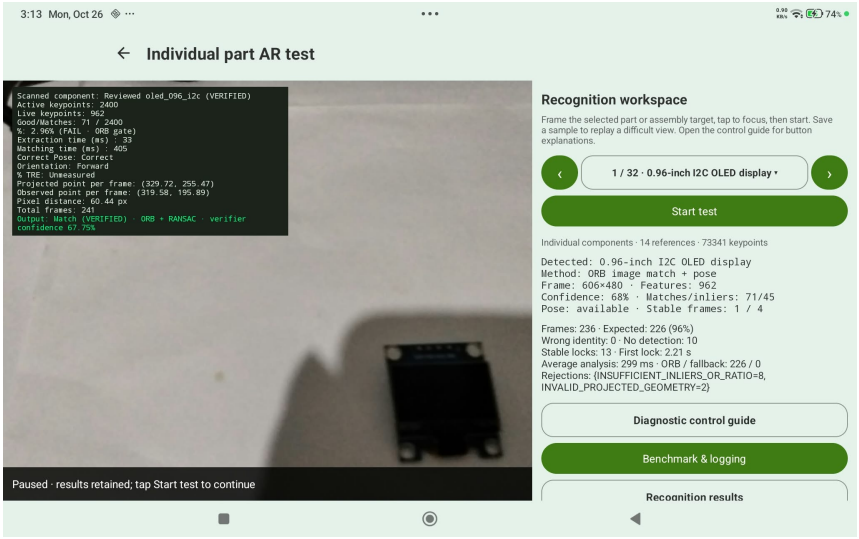



Figure 1. User-supplied paused OLED test; the report transcribes its displayed values.

## 7. Session results measure detection frequency

**226 / 236 tester frames = 95.76%, displayed as 96%.** This is the frequency of expected detections in this run, not ORB match yield or independently validated classification accuracy. The session shows 10 no-detection frames, 0 wrong-identity frames, 13 stable locks and a first lock at 2.21 seconds.

**Stable frames: 1/4** means the displayed observation had not completed a new stability streak; earlier locks remain counted. **Total frames: 241** is the cumulative analyzer counter, whereas 236 belongs to this tester result group. Pause retained results; it did not convert the 85% gate into a pass.

## 8. What this view does and does not prove

The selected OLED was recognized repeatedly, and the displayed frame had spatially verified ORB evidence. The separate 85% yield criterion failed. This view does not establish a calibrated recognition probability, dataset-wide accuracy, physical seating or electrical continuity.

TRE remains Unmeasured without independent metric ground truth. The **60.44 px** center-to-inlier-centroid distance compares different point definitions; it is not solver RMS reprojection error. Extraction **33 ms** and matching **405 ms** describe the displayed frame, not the full human recognition task.

## 9. Reporting recommendation - no code change

Report **match yield**, **RANSAC inlier fraction**, **recognition outcome** and **session detection rate** separately. The requested 85% yield rule is not a validated universal recognition-success criterion. Do not change counts, denominators or thresholds merely to obtain PASS; justify any future protocol revision using labeled positive and negative trials.

Basis: supplied screenshot and read-only inspection of DiagnosticScanTelemetry, DiagnosticHammingMatcher, OrbMatchPolicy, ComponentRecognizer and VisionConfig. No application code or scan thresholds changed.